Reading & Remarks

Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits 5th Edition*, McGrawHill, 2015.

- Chapter 4

Remarks:

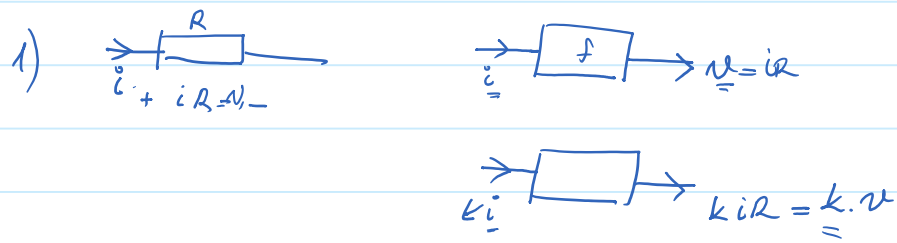
- Quiz - 1 ✓
- Homework - 2 ✓

Linearity

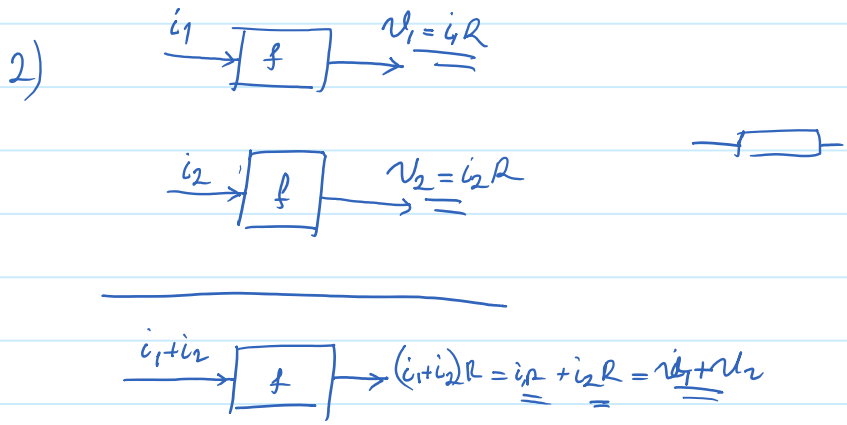
1. Homogeneity
2. Additivity

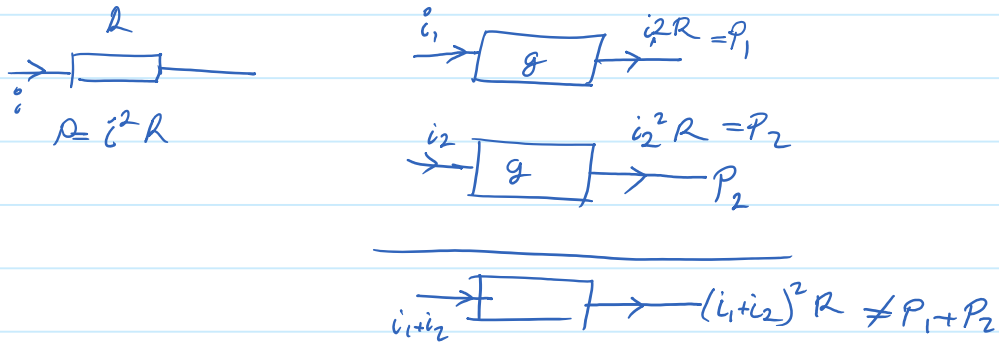
A resistor is linear. Relation between input current and output power is not linear.

1)



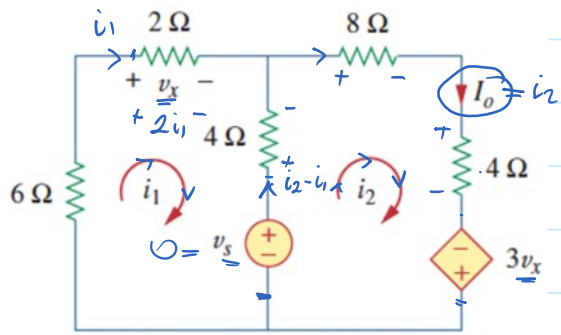
2)





Exercise

For the circuit below, find I_0 when v_s = 12V and v_s = 24V



$$-v_s - 6i_1 - 2i_1 - 4(i_1 - i_2) = 0$$

$$v_s - 4(i_2 - i_1) - 8i_2 - 4i_2 + 6i_1 = 0$$

5/ $-12i_1 + 4i_2 = 12 = v_s$ (24)

6/ $10i_1 - 16i_2 = -12 = -v_s$ (-24)

$$-60i_1 + 20i_2 = 60$$

$$60i_1 - 96i_2 = -72$$

$$-76i_2 = -12$$

$$i_2 = \frac{-12}{-76} = \frac{6}{38} = \frac{3}{19}$$

$v_s = 12V$

$v_s = 24$

$$-76i_2 = -24$$

$$i_2 = \frac{12}{38} = \frac{6}{19}$$

$v_s = 0 \rightarrow i_2 = 0$

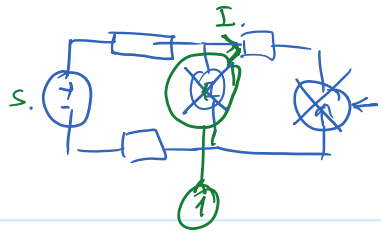
×2

×2

$$I_o = i_2 = \frac{12}{76}\text{A}$$

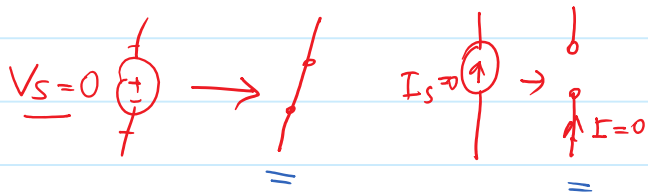
$$I_o = i_2 = \frac{24}{76}\text{A}$$

Superposition

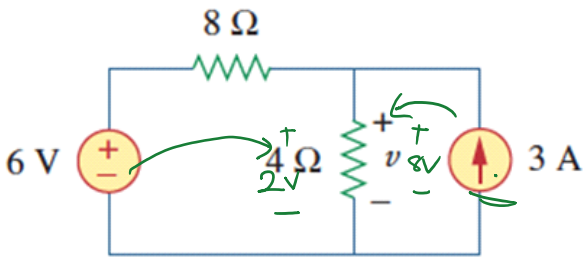


$I_1 + I_2 + I_3 = I$

1) Turn off all independent sources
   -> replace voltage source with 0V (short circuit)
   -> replace current source with 0A (open circuit)
   Find the output (voltage or current) due to that active source
2) Repeat step 1 for all independent sources
3) Find the total contribution by adding algebraically all the contributions due to independent sources

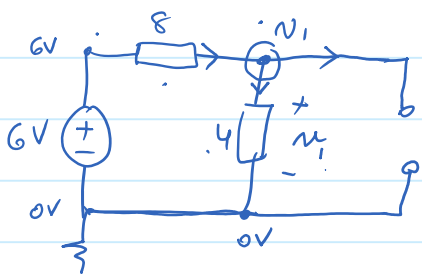


Exercise

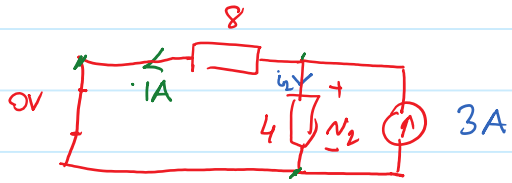


$v = v_1 + v_2$

$= 2 + 8 = 10V$



$v_1 = \frac{6}{8+4} \cdot 4 = 2V$



$i_2 = \frac{3}{8+4} \cdot 8 = 2A$

$v_2 = i_2 \cdot 4 = 2 \cdot 4 = 8V$

$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Exercise

1 5 _ _ _ 1234 1/3

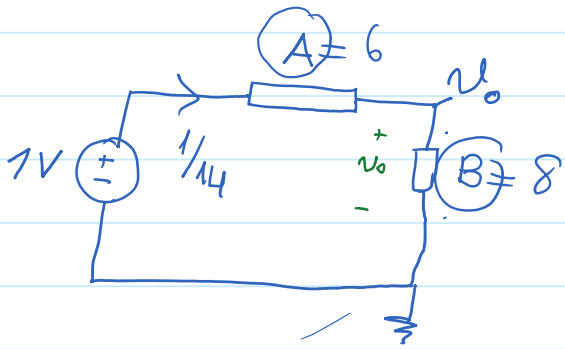
1pts **Q1*** Last four digits of your student ID

Example: 21 → 1 2 3 4 → 1+2+3 = A=6, 3+4+1 = B=8

Answer: 68

* if not correct no credits

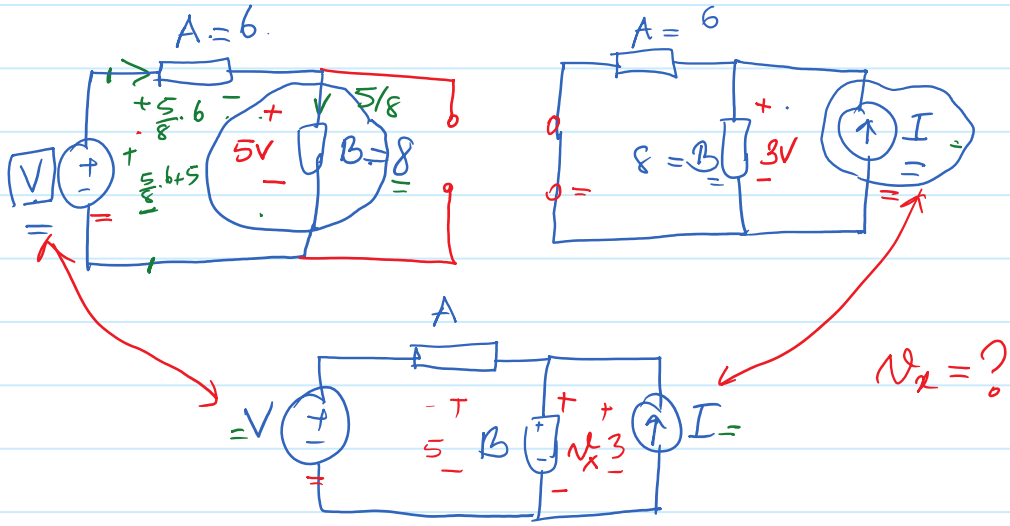
1pts **Q2.**



$v_o = ?$

$8 \cdot \frac{1}{14} = \frac{8}{14}$ (1)

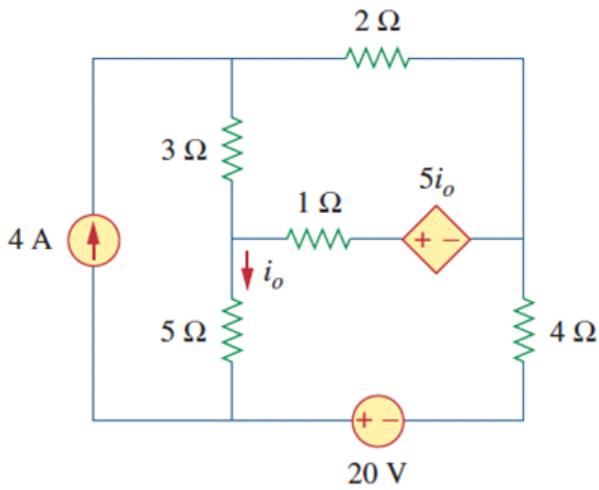
1pts **Q3.**



Hint: Use superposition.

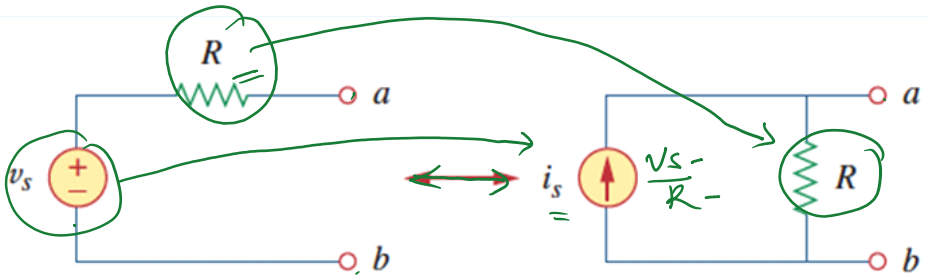
$v_x = 5 + 3 = 8$

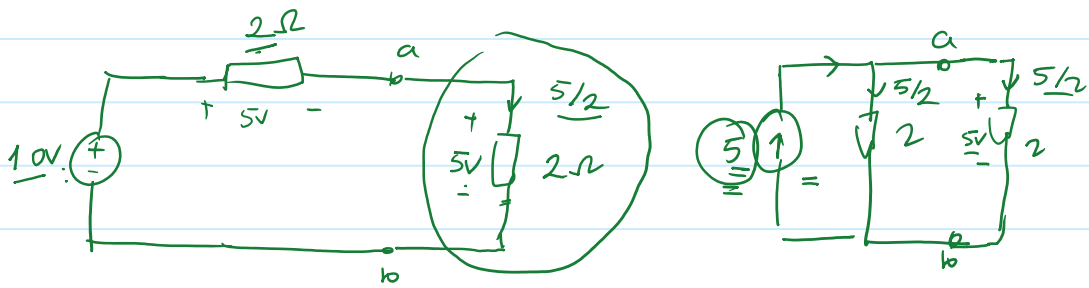
Exercise



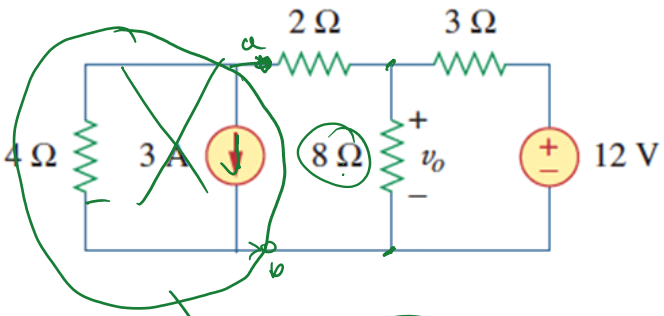
$$i_o = -\frac{8}{17} = -0.4706 \text{ A}$$

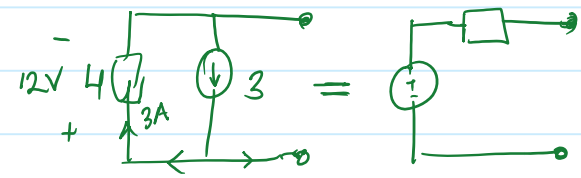
Source Transformation

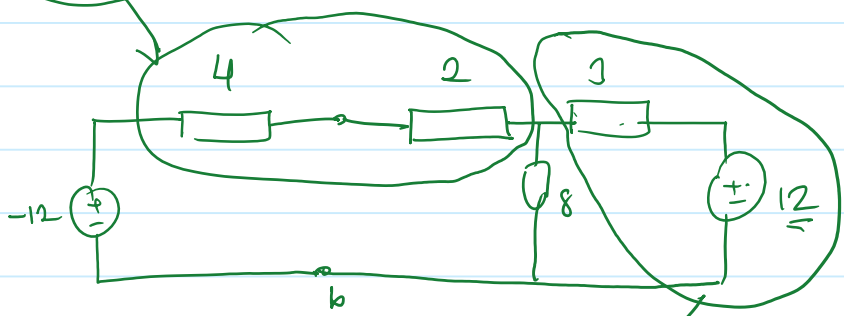


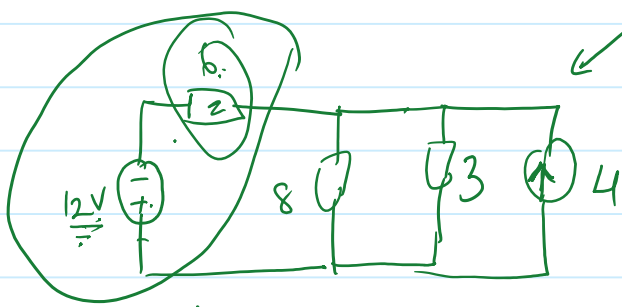


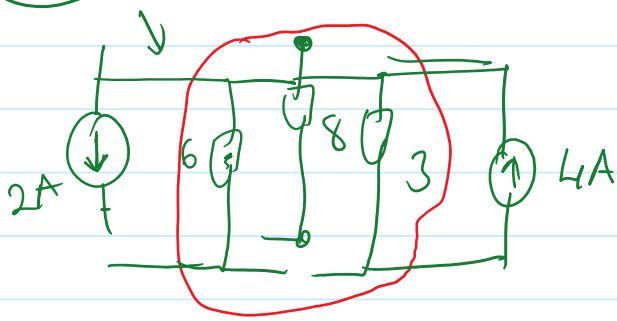
Exercise





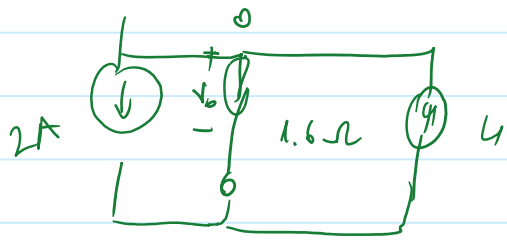


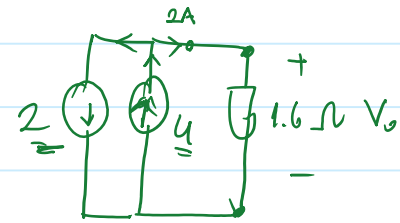


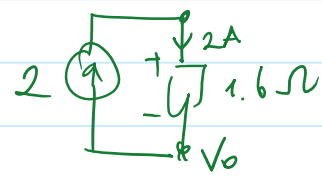


$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{8} + \frac{1}{3}$$

$$R_{eq} = 1.6\ \Omega$$

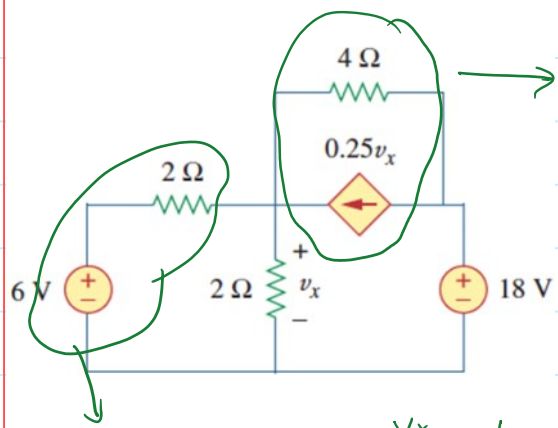




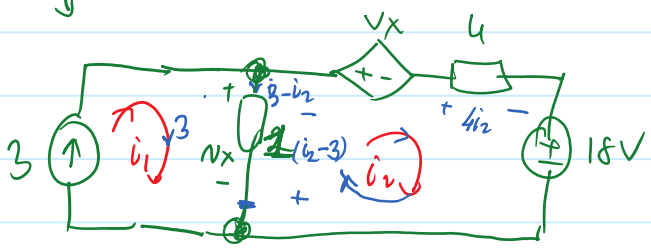


$1.6 \cdot 2A = 3.2V$

Exercise







$$v_x = (3 - i_2) = 3 - \left(-\frac{9}{2}\right) = \frac{15}{2}$$

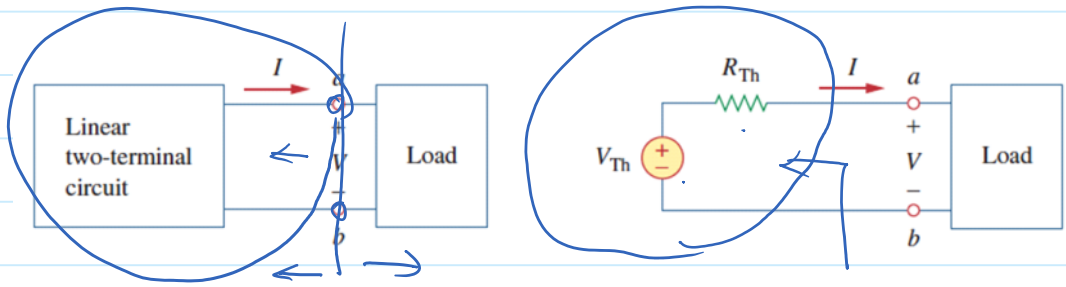
$i_1 = 3A$

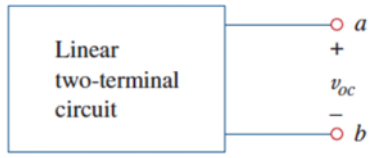
$$-(i_2 - 3)1 - v_x - 4i_2 - 18 = 0$$

$$-i_2 + 3 - 3 + i_2 - 4i_2 - 18 = 0$$

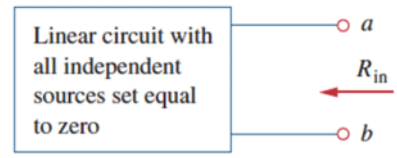
$$i_2 = -\frac{18}{4} = -\frac{9}{2}$$

Thevenin's Theorem



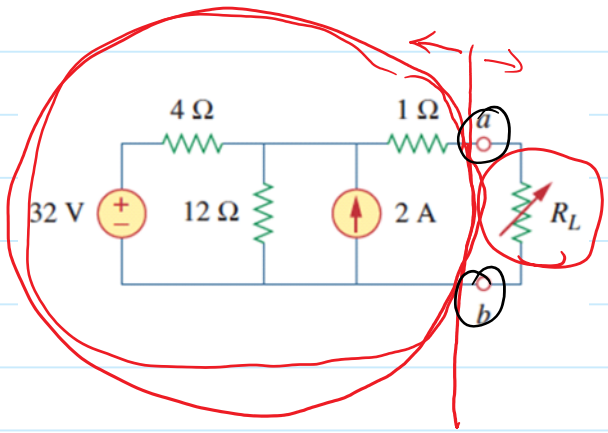


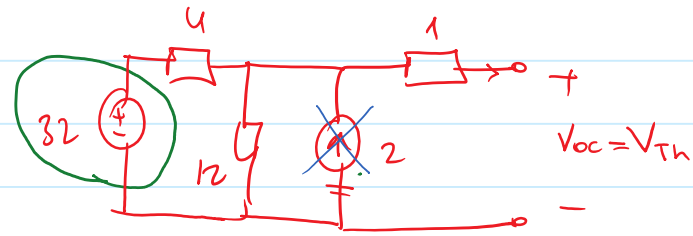
$V_{Th} = v_{oc}$

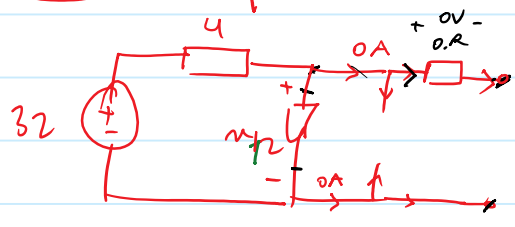


$R_{Th} = R_{in}$

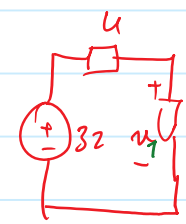
Exercise





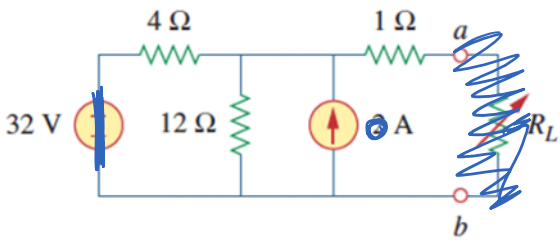


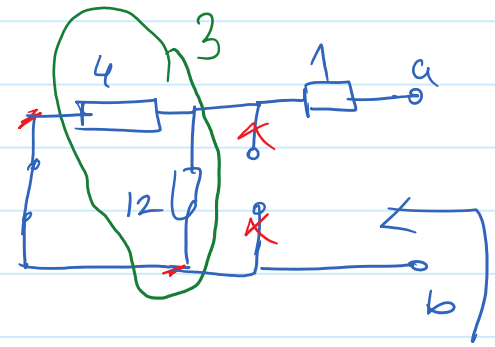
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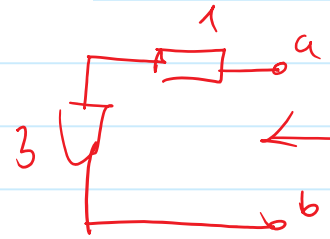


$v_1 = \frac{32}{16} \cdot 12 = 24V$

$V_{oc} = 24V = V_T$ $\qquad R_{Th}$

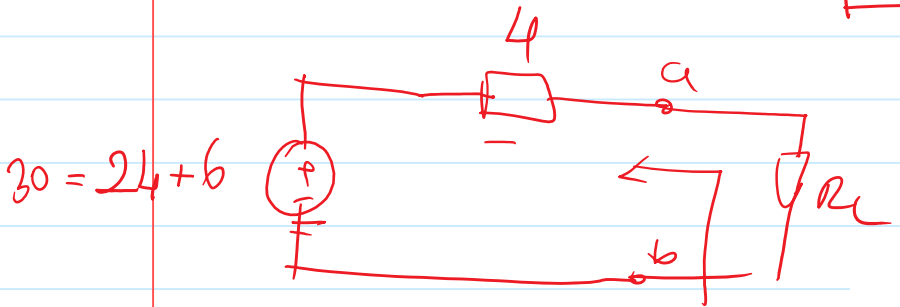




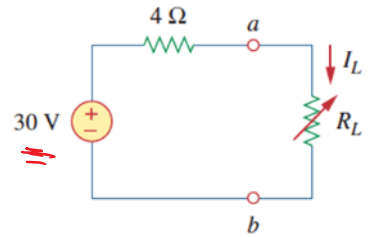


$R_{Th} = R_{eq}$

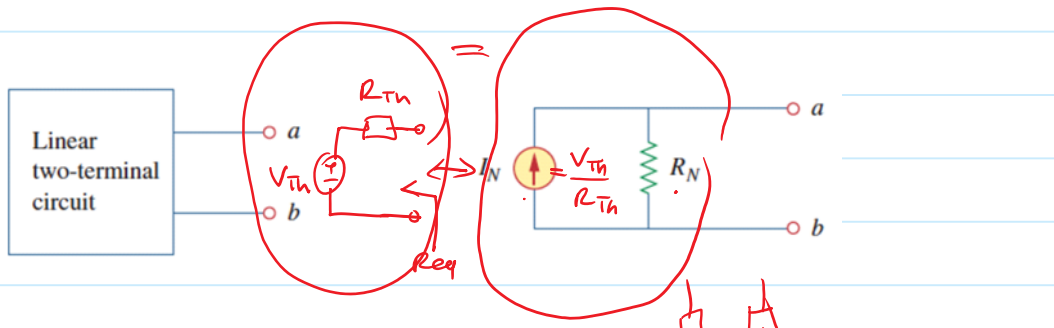
$R_{eq} = 4 = R_{Th}$



$30 = 24 + 6$

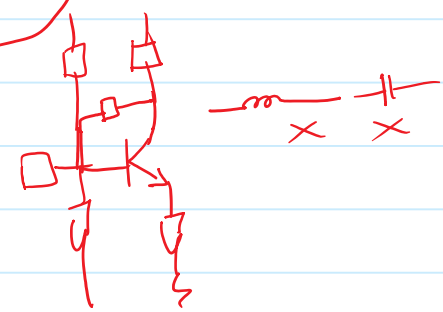


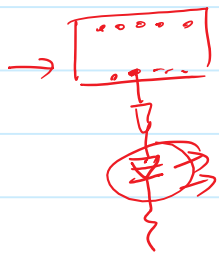
Norton's Theorem

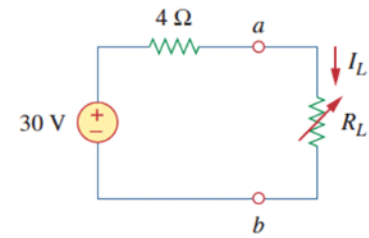


$$R_N = R_{\text{Th}}$$

$$I_N = \frac{V_{\text{Th}}}{R_{\text{Th}}}$$






4 Ω
a
I_L
30 V
R_L
b